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Global supply chains and cross-border financing

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Abstract

This paper provides evidence that the formation of global supply chain partnerships leads to an increased usage of cross-border financing. The findings are detected in all three major financing markets—equities, syndicated loans, and public debt. Difference-in-differences tests allow us to draw a causal interpretation of our main findings, which also holds in several robustness tests. Our findings suggest that increased cross-border financing reflects greater ability to access global financial markets, due to enhanced firm visibility and investor attention, as well as operational forces that arise when establishing new global supply chain partnerships. Specifically, we provide evidence that firms that are small, held less by institutional investors, and followed by fewer analysts have more benefits in cross-border financing from the formation of the global supply chain. Our findings have important implications for firms attempting to integrate into the global supply chain network. More broadly, our findings suggest that the information generated by operational activities can have significant effects on subsequent financing activities.

KEYWORDS
cross-border financing, global supply chains

1 | INTRODUCTION

Two important components of economic globalization are supply chain globalization and financial globalization (IMF, 2002). Supply chain globalization refers to the global linkages that result from cross-border procurements of goods or services. It offers firms the advantages of cost-savings (Lai et al., 2021; C. Li, 2013; Shao et al., 2020), access to materials or production capabilities that may be unavailable domestically (Jain et al., 2013), new technology, and specialized knowledge (Berry & Kaul, 2015; Gray & Meister, 2004), etc. Financial globalization refers to global linkages that arise through cross-border financial flows. Theory and empirical evidence suggest that financial globalization promotes diversification benefits for investors (French & Poterba, 1991), lower financing costs for firms (Doidge et al., 2004), and better allocation of the world's supply of capital (Feldstein & Horioka, 1980; Lucas, 1990).

While supply chain globalization is relatively well developed, financial globalization is less complete, primarily due to transaction costs and informational frictions that impede securities trading and capital raising. Prior literature identifies information asymmetry as a major obstacle to cross-border financing (CBF) (Erel et al., 2012; Stulz, 2005). While these studies mostly focus on variation in information asymmetry aggregated at the country level, an open question is to what extent firms can independently overcome these obstacles to obtaining CBF.

In this paper, we investigate the relationship between global supply chain (GSC) formation and subsequent (CBF). We consider two nonmutually exclusive potential reasons for why firms engage in more CBF after integrating into the GSC network. First, we propose that by integrating into the GSC network, a firm may be able to overcome informational obstacles to CBF. In particular, establishing a GSC relationship can raise the visibility of the partnership, attract investor attention, and convey an informational benefit as a corporate partner abroad conducts due diligence before forming a GSC relationship (Cen et al., 2015; Honhon et al., 2012; Merton, 1987). The seminal paper of Merton (1987) presents the *investor recognition hypothesis*, predicting that investor attention can alleviate informational frictions that prevent them from holding lesser known assets. Cen et al. (2015) consider the investor recognition hypothesis in a supply chain setting and find that banks perceive suppliers as less risky

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All authors have contributed equally to this study and are listed alphabetically.

if they have reputational customers and thus can borrow at lower rates with loan covenants that are less strict. Therefore, a new GSC relationship may reduce information asymmetry and provide favorable information regarding a firm's products and operations as well as its financial health. Moreover, since a GSC relationship is less affected by nonmarket forces such as local cronyism, it also helps to reduce opaqueness and uncertainty for global investors.

In addition to greater access due to increased informational visibility, there are operational reasons why firms may engage in more CBF after integrating into the GSC network. For example, firms may use CBF to hedge increased currency risk that may have arisen from globalizing their supply chain network. There are also potentially important product feedback market effects that can arise from cross-border capital raising. For example, firms may be able to glean information about product markets when negotiating CBF deals with banks and underwriters. Our findings suggest that both increased access and operational forces explain increases in CBF following the formation of new GSC relationships.

To investigate the relationship between GSC formation and subsequent CBF, we begin by assembling a large sample of GSC relationships and CBF deals. Our combined dataset is novel, in large part, because a lack of common codings for firms across the requisite datasets makes merging global financial markets data and GSC data difficult. We carry out an endeavor of manual name matching with careful cross-checking that merges information on GSCs from FactSet Revere, financial information from Worldscope, cross-border bond issuances from Datastream, equity cross-listings from major depositary-receipt banks and stock exchanges, and global syndicated loans from DealScan. In our final regression sample of 343,547 firm-year observations, we manage to match 22,199 CBF deals, resulting in 11,715 firm-year observations with at least one CBF deal.

Our empirical investigation begins with a panel analysis that shows, after controlling for firm characteristics and using fixed-effect specifications, firms with more GSC relationships have a higher likelihood of obtaining CBF. The evidence is robust across the three major financing markets—equities, syndicated bank loans, and public debt.

Next, we employ a difference-in-differences (DID) analysis with matching that shows that the formation of GSC relationships is associated with a sustained increase in CBF. We match on ex ante firm attributes and whether the firm carried out any CBF deals in the 3 years prior to establishing its first GSC partnership. In parallel trend analysis, we confirm that the treated and control firms have similar trends in CBF in the pre-event period. The increase in CBF is robustly found across the three major financing markets we study. Aggregating across all three markets, our estimates show that when a firm switches from having no GSC relationships to having at least one, there is an increase in the probability of using CBF of 1.7 percentage points, which is a 59% increase compared to the annual probability of a CBF deal. The increase in the probability of CBF is most substantial in the bond market (1.2 percentage points, a 50% increase), followed by syndicated loans (0.4 percentage points, an 87% increase) and cross-border equity issues (0.1 percentage points, a 157% increase). Our results also hold across multiple robustness tests, including alternative CBF measures, alternative model specifications, and when we use exogenous trade agreements as instrumental variables (IV) to enhance the causal interpretation of our main findings.

Finally, we investigate the cross-sectional nature of our sample to further characterize the effect of establishing new GSC relationships on subsequent CBF and to bolster support for the validity of our conclusions. Since a prerequisite for raising funds for operational reasons is the ability to access global financial markets, these tests focus largely on factors that identify heterogeneity across firms in the benefit of establishing new GSC relationships in lowering information obstacles to obtaining CBF. We expect firms with lower visibility to receive more marginal benefits from establishing new GSC relationships. We utilize three empirical proxies of information visibility that are frequently used in the visibility literature: firm size (Bushee & Miller, 2012), institutional ownership (Han et al., 2017), and the number of analysts following the firm (O'Brien & Bhushan, 1990; O'Brien & Tan, 2015). We find a significantly larger increase in CBF for firms that are small, have lower institutional ownership, and are followed by fewer analysts. Therefore, less visible firms indeed benefit more in terms of using CBF after GSC integration.

Our paper contributes to several strands of literature. First, we contribute to the growing empirical supply chain literature (Cai et al., 2014; Croson et al., 2014; Davies & Joglekar, 2013; Huo et al., 2014; Jain & Wu, 2023; Kleindorfer & Saad, 2005; Osadchiy et al., 2021; Wang et al., 2021). While most of the extant empirical studies focus on domestic supply chain relationships (Bray et al., 2019; Hertzel et al., 2008; Johnson & Whang, 2002; Lee et al., 2017; X. Li et al., 2016; Osadchiy et al., 2015), our study is among the first to extend the scope to GSC relationships. Although supply chain relationships and financial globalization have been widely studied individually, to the best of our knowledge, the relationship between firm-level decisions on global goods and service market transactions and CBF has not been empirically addressed. Using diverse identification strategies, we establish a causal interpretation between the establishment of GSC relationships and subsequent CBF deals. Ours is the first study to examine the relationship between these two key components of economic globalization at the firm-pair level.

We also contribute to the literature that examines financing along the supply chain. This literature has largely focused on financing *within* supply chain relationships, that is, supplier-led financing (Chod et al., 2019; Kouvelis & Zhao, 2012; Tunca & Zhu, 2017; Yang et al., 2021) or buyer-led financing (Chen et al., 2020; Tang et al., 2017). All else equal, this type of *internal* financing is likely to increase mechanically as new supply chain partnerships are established. In contrast, our paper focuses on capital raising in financial markets, which are *external* to the supply chain relationship and, therefore, should not necessarily lead to mechanical increases in supply chain financing. Thus, our study expands the traditional view of supply chain financing and provides novel evidence on the use and determinants of CBF that should be of interest to academics and practitioners.

Taken together, our results also provide useful managerial insights, especially for those managers tasked with identifying supply chain partners and deciding whether to integrate into the GSC network. Our findings suggest that increased access to global financial markets, as well as operational aspects of CBF, should be important considerations when determining whether to form a GSC partnership. The benefits of increased usage of CBF extend beyond the cost of capital to the added security bought about by the diversification of the firm's capital sources. Potential operational benefits from CBF include knowledge gained from interactions with global financial intermediaries, improved governance, and certification effects from broader capital market monitoring, possible product market spillover or feedback effects, and the ability to hedge foreign currency risk. Furthermore, our heterogeneity analysis provides managerial guidance as to when the establishment of GSC partnerships is more likely to lead to improved usage of CBF. Finally, as a practical matter, our findings suggest that supply chain managers should coordinate with their CFOs, who should also have a direct interest and useful guidance on decisions on forming new GSCs.

The remainder of the paper proceeds as follows. Section 2 describes our data and sample construction using automatic and manual textual matching. Section 3 develops our main hypothesis and specifications, reports the empirical results using panel regression, and then difference-in-differences with matching. Section 4 presents additional tests to verify the robustness of our main findings. Section 5 presents the cross-sectional analyses. Section 6 concludes.

2 | DATA DESCRIPTION AND SAMPLE CONSTRUCTION

We assemble a comprehensive dataset of GSC relationships and CBF deals in the three major global financial markets (corporate bonds, common stock, and syndicated loans). The primary universe of our study is all global public firms from the Thomson Reuters Worldscope database, where we obtain annual financial variables. Specifically, global firm-level supply chain and CBF data are described below.

2.1 | Data sources for GSC and CBF and summary information

Our GSC data are from FactSet Revere between 2003 and 2015. FactSet hand-collects and verifies supply chain information from a wide range of sources in different languages globally, including public filings, sell-side presentations, press releases, firm websites, product catalogs, and major news media. FactSet monitors company 10-K filings, investor presentations, and websites annually, while company press releases and corporate actions are monitored daily. Recent studies that use FactSet Revere supply chain data include Modi and Cantor (2021), Osadchiy et al. (2021), Wang et al. (2021), Agca et al. (2022), Wu et al. (2022), Charoenwong et al. (2023), and Ding et al. (2023), etc.

Table 1 displays the top 40 countries or regions with the most GSC relationships. The first column shows the country name. The second column shows the total number of unique firms involved in GSC networks in that country. The other cells in the lower triangle of the matrix represent the number of unique GSC observations involving two firms, one in the column country and the other in the row country. For example, in the second row, we can observe that our sample captures 949 public firms in Japan, and there are 9085 unique supply chain relationships between Japan and the United States. The United States has the largest number of companies with GSC coverage, followed by Japan, the United Kingdom, and Germany. This table shows that our sample has significant coverage of supply chain relationships for all major countries.

Our CBF observations are assembled from a battery of different sources for major financial markets: bond, equity, and syndicated loans. A corporate bond is a nonintermediated debt between investors and a corporation. Our data on cross-border bond issuances come from Thomson Reuters Datastream, where we collect 86,975 global bond issues. Panel A of Figure 1 shows the distribution of global bond issues by the home countries of issuing firms. The United States ranks first as it has 30% of all global bond issuance deals. It is followed by the United Kingdom (9%), China (8%), and the Netherlands (7%). There are 17 countries with more than 1% of all deals, showing a broad market in terms of the home country of issuing firms.

Cross-listing is when a firm issues its equity shares on a foreign stock exchange or sells its shares as bank depositary receipts. We focus on capital-raising cross-listings, ones through which firms obtain CBF, by issuing shares after cross-listing. We hand-collect 1449 cross-listings from major depositary receipt banks (BNY-Mellon, JP Morgan, Citi, Deutsche Bank, etc.) and exchanges (NYSE and NASDAQ). Panel B of Figure 1 shows the distribution of home countries of firms in these equity cross-listing deals. Notably different from global bond issues, China has the largest number of cross-listings (23%), followed by the United Kingdom (9%), Australia (7%), and Russia (7%). There are 21 countries with more than 1% of all deals.

Global loans are bank loans from a foreign bank or a group (syndicate) of lenders led by a foreign bank. We collect 10,104 global loans from Thomson Reuters DealScan. Panel C of Figure 1 shows the distribution of global loans for firms in each country. This is a more segmented market as none of the countries alone has more than 15% of the global market. The United States has the largest number of transactions

TABLE 1 Global supply chain coverage.

ticker	country name	firm	US	JP	GB	DE	FR	CA	KR	CH	HK	TW	SE	IT	AU	CN	IN	BR	NL	MX	SG	ES	ID	NO	IL	CL	ZA	BE	FI	DK	TR	RU	MY	TH	PL	AR	AT	GR	NZ	PH	IE
US	United States	3772																																							
JP	Japan	949	9085																																						
GB	United Kingdom	814	7422	545																																					
DE	Germany	808	5849	394	632																																				
FR	France	604	4407	367	691	563																																			
CA	Canada	471	4110	232	379	250	222																																		
KR	South Korea	456	2175	479	175	145	147	66																																	
CH	Switzerland	452	2141	174	213	269	189	73	59																																
HK	Hong Kong, China	394	1305	260	205	151	155	60	87	71																															
TW	Taiwan, ROC	390	2319	485	99	140	51	27	171	46	64																														
SE	Sweden	378	1414	152	214	207	172	55	61	88	50	19																													
IT	Italy	303	1313	140	199	207	260	74	52	64	55	17	55																												
AU	Australia	213	1255	170	319	68	88	180	46	39	66	21	30	28																											
CN	China	212	717	208	50	147	69	27	112	25	1066	110	29	25	21																										
IN	India	194	1156	137	242	144	131	71	67	67	37	24	66	68	38	27																									
BR	Brazil	179	1160	123	127	168	204	51	37	41	11	5	54	87	22	7	14																								
NL	Netherlands	159	1138	98	165	178	175	45	33	28	31	39	68	45	28	23	26	68																							
MX	Mexico	142	979	104	82	80	80	43	28	39	4		23	30	7	4	4	45	37																						
SG	Singapore	142	636	130	108	66	71	22	41	23	77	19	25	11	73	36	22	18	19	8																					
ES	Spain	134	611	35	142	92	121	48	27	31	19	5	35	50	18	4	16	58	36	26	13																				
ID	Indonesia	131	419	393	75	74	91	22	47	33	35	20	24	13	37	21	29	1	21		75	8																			
NO	Norway	122	509	34	155	83	85	47	38	19	8	6	99	32	35	13	27	34	17	1	41	21	6																		
IL	Israel	119	819	37	66	67	62	41	16	26	4	8	30	24	4	6	11	5	11		5	3																			
CL	Chile	115	415	64	134	63	71	26	29	28	5	4	10	9	28	8	9	46	32	33	3	80	1	14	2																
ZA	South Africa	110	382	77	119	66	40	24	15	14	1	10	18	26	78	5	18	12	17		4	5	2	5	1																
BE	Belgium	107	446	41	64	61	93	11	18	28	11	4	13	7	10	2	6	42	9	11	3	8	6	6	4	16	4														
FI	Finland	104	229	23	60	51	38	20	16	12	6	3	105	8	16	11	8	24	13	8	5	7	11	18	7	8	7	6													
DK	Denmark	96	297	31	57	38	36	27	19	10	10		52	9	11	19	19	13	12	3	10	8	4	23	9	10		6	16												
TR	Turkey	89	252	68	66	84	47	4	20	22	8	9	24	35	3	1	4	2	13		2	11	4	4	1	2	4	8	3	2											
RU	Russian Federation	89	203	37	141	69	49	8	24	7	9	1	10	19	5	7	7	5	12	1	3	4		11	1	2	1	2	17	7	4										
MY	Malaysia	72	192	86	60	28	30	24	21	7	22	10	7	11	14	6	20	3	10		46	10	25	11				2	5	5	3	2									
TH	Thailand	66	230	98	40	36	25	11	24	8	10	16	7	4	9	5	13		5	4	17	3	20	8	1	1	1	6	4	1	3	2	20								
PL	Poland	64	156	17	42	65	69	12	6	17	3	5	24	18	2	2	5		15		1	11		4	3	2	1	5	16	3	5	10	1								
AR	Argentina	62	206	6	29	31	58	14	10	4		1	8	16	2	1	1	62	3	3		37		2	1	49	1	3													
AT	Austria	56	106	10	33	61	26	6	8	15	8	4	20	12	8	16	8	8	13	6	1	8	2	9	2	3	7	2	7	1	15	7	1	4	17						
GR	Greece	54	128	11	46	31	15	3	4	9	4	2	7	7	3	2	3		12		2	6	1			2	3	3	1	8	2	1			1		7				
NZ	New Zealand	53	116	16	33	10	15	12	3	11	6	1	2		67	4		1	2	1	7					1	3										1				
PH	Philippines	46	111	32	17	7	14	10	5	4	13	1	5	1	11	1	6	2	4	3	12	4	12		3	1	2		3	4			1	10					1		
IE	Ireland	45	188		31	10	13	14		7			2	2	1		1	1	5	1		1	1	1				7		4					1						
PE	Peru	43	77	17	37	9	13	5	2	6	2		5	2				13		9	2	12	1			52	1	4								3					

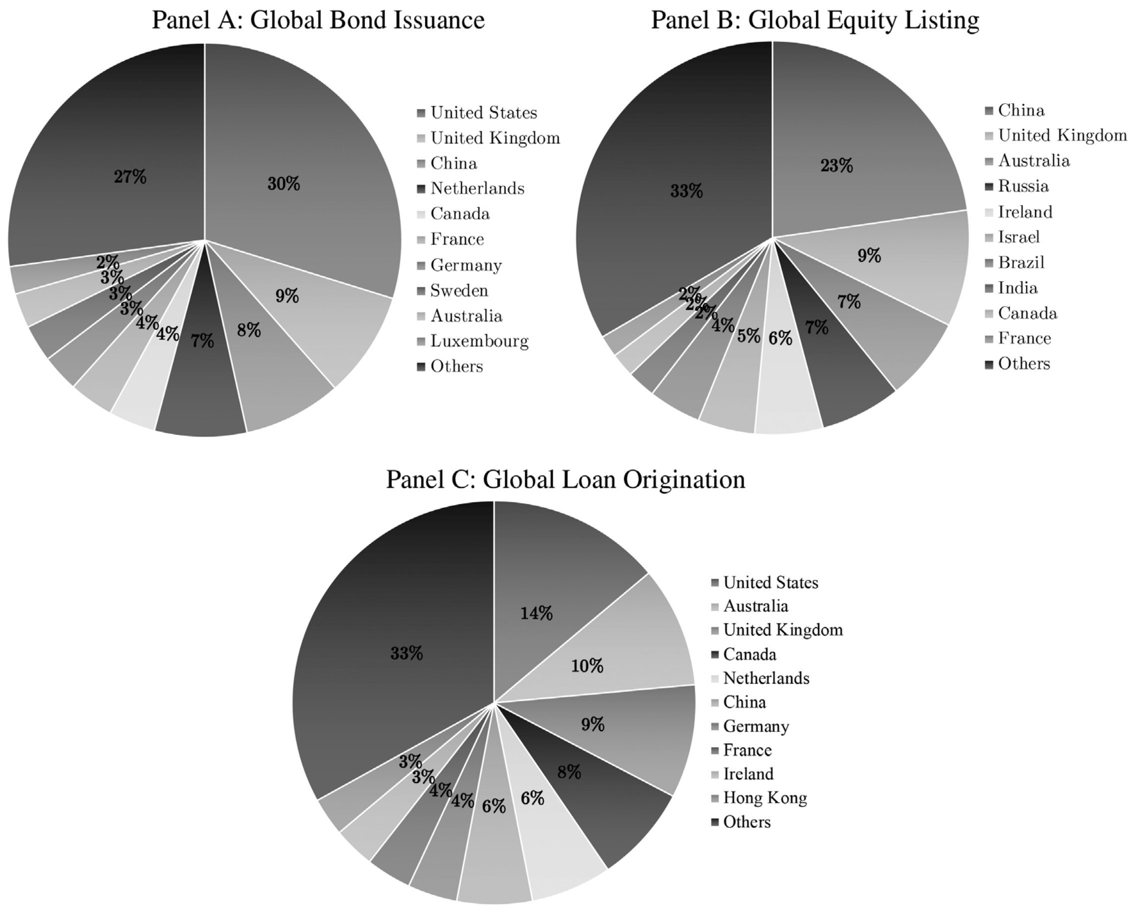

FIGURE 1 Global CBF distribution by country.

(14%), followed by Australia (10%) and the United Kingdom (9%).

2.2 | Sample construction

Since the CBF datasets lack a unified firm identification code system at the global level, we use name-matching algorithms and manual checks with cross-validation to construct our sample. The steps for identifying our sample are shown in Appendix Figure A1 in the Supporting Information. To begin, we combine firm-level fundamental information from Worldscope with GSC relationship data from FactSet Revere. We then merge this firm-level fundamental and GSC information with CBF deals identified from different sources by fuzzy name matching.[1] In this way, we create a unique merged dataset containing GSC relationships as well as the global financial market at the firm level. In our final regression sample of 343,547 firm-year observations, we manage to match 22,199 CBF deals, resulting in 11,715 firm-year observations with at least one CBF deal.[2] We find that the number of deals linked via the manual matching procedure contributes 23.46% of the number of all deals linked through exact and manual matching, suggesting our endeavor of manual name matching is effective in enabling the data sample for the empirical analysis.

3 | EMPIRICAL ANALYSIS: GSC FORMATION AND CBF USAGE

In global financial markets where the presence of informational frictions is common, the establishment of GSC relationships may facilitate firms' usage of CBF by providing valuable information on financial health and product quality as well as by increasing firm visibility and attracting investor attention. In addition, CBF can result from operational needs that may arise when forming a new GSC relationship. We proceed in two steps to test whether GSC formation increases subsequent usage of CBF. First, we use a set of panel regression specifications to provide preliminary evidence. In the second step, we use a difference-in-differences strategy with matching on firm attributes and CBF outcomes prior to GSC establishment to alleviate the concern that the panel regression estimates may be purely due to differential pretrends and to estimate the dynamics of CBF outcomes once the firm has established its first GSC relationship.

3.1 | Panel regression

To provide preliminary evidence on the effect of GSC formation on subsequent CBF activity, we employ the following panel regression specification:

$$CBF_{it} = \beta \log(1 + NGSC_{it}) + \gamma X_{it} + \delta Z_{kt} + \phi_k + \chi_j + \psi_t + \epsilon_{it}, \quad (1)$$

where CBF_{it} is a dummy variable indicating whether firm i in industry j and country k is involved in any CBF deals in year t. $NGSC_{it}$ is the number of GSC relationships firm i has in year t. We take the logarithm of $NGSC_{it}$ plus one to reflect that the average increase in CBF associated with establishing one more GSC relationship is unlikely to be the same when the firm already has, for example, 1 versus 10 GSC relationships, and to ensure the estimation is not overly sensitive to observations with outlying values in $NGSC_{it}$.[3]

Furthermore, we include a battery of control variables for firm-level characteristics (ROA_{it}, $Sales\ Growth_{it}$ eqnarray, $\log(Sales)_{it}$, and $Leverage_{it}$), country-level attributes ($Financial\ Development_{kt}$ and $Financial\ Openness_{kt}$), and institutional quality measures ($Voice\ and\ Accountability_{kt}$, $Political\ Stability_{kt}$, $Government\ Effectivene\ ss_{kt}$, $Regulatory\ Quality_{kt}$, $Rule\ of\ Law_{kt}$, and $Control\ of\ Corruption_{kt}$), represented by the vector X_{it} and Z_{kt}. *Financial Development* is measured as the ratio of the sum of domestic credit outstanding and equity market capitalization to the gross domestic product (GDP). *Financial Openness* is measured by the capital account openness index in Fernández et al. (2016). In all regressions, we control for country, industry, and year fixed effects, and use clustered standard errors at the firm level, which addresses heteroskedasticity and autocorrelation. Appendix Table A1 in the Supporting Information presents the definitions and sources for all variables. All control variables are standardized to have a mean of zero and unit variance.

Table 2 presents the panel regression results, which are consistent with the prediction that GSC relationships have a significant impact on firms' global bond issuances, global equity listings, global loan originations, as well as cross-border deals in the three markets combined. The main explanatory variable of interest is the number of GSC relationships measured in log numbers. We find significant associations between the number of GSC relationships and CBF deals: Going from zero to one GSC relationship is associated with an increase in the probability of using CBF of 1.89 percentage points (calculated by multiplying an increase of 0.7 points in the independent variable, the logarithm of one plus the number of GSC relationships, by the estimated coefficient of 0.027). This suggests that a firm's decision on global goods and services transactions may matter for CBF, over and beyond country and industry factors.

3.2 | Difference-in-differences with matching

One concern with the preliminary panel regression is that the coefficient on GSC relationships might only be picking up differential pretrends for different firms. For example, it is possible that the coefficient on GSC relationships is entirely driven by an omitted variable problem, which firms participating in GSC relationships, and the remaining firms may have diverging trends in using CBF, potentially due to different productivity levels or growth trajectories. To address this concern, we estimate a difference-in-differences model with matching on firm attributes and CBF outcomes prior to GSC establishment. The assumption of the DID analysis is that by careful matching, the trend in CBF of the control firms provides an adequate proxy for the trend in CBF of the treatment group in the absence of treatment, that is, in the absence of forming a GSC relationship. Our main hypothesis predicts that by comparing two firms that are otherwise similar in ex ante outcome and attributes, we should observe more CBF for firms that establish GSC relationships.

3.2.1 | Balance tests and estimation results

For the DID analysis, since a firm's size (which we proxy by sales), growth, profitability, and financial structure (which we proxy by leverage) are the important factors for integrating into the global value chain in the international trade literature (Bernard & Jensen, 1999; Delgado et al., 2002), we use Mahalanobis distance matching to identify the control group based on sales, sales growth, return on assets, and leverage over the 3 years prior to GSC formation. We also match on a dummy variable ($PreEvent\ \ 1\{CBF_{it}\}$) indicating whether the firm has been involved in any CBF deal in the prior 3 years to hold fixed usage of CBF before the treatment firm establishes its first GSC relationship, as well as on the firm's country and industry using Fama-French industry classification[4] based on four-digit standard industry classification codes to hold fixed the economy-wide and industry-wide factors. These variables are all available in our dataset with good coverage across firms.

For each treated observation, we match one controlled observation. We include only observations in the [−3,5] year time interval around GSC formation in the DID estimation sample. The DID sample is hence a smaller sample than the panel regression sample. The total number of observations in the DID sample is 92,046, about one-fourth of the panel regression sample. Naturally, the number of CBF deals that occurred in the DID sample is also smaller in total—we observe 4501 CBF deals in the DID sample across 2651 firm-year observations.[5]

In Table 3, we report balance tests to confirm the effectiveness of our matching. *Diff*. is calculated as the difference in the mean values for the treatments and controls. As shown in Table 3, all targeted characteristics of the treated and control observations are well balanced, as evidenced by the

TABLE 2 Panel regression results.

Markets	Global bond issuance (1)	Global equity listing (2)	Global loan origination (3)	Cross-border financing (4)
log(1 + *NGSC*)	0.021***	0.002***	0.005***	0.027***
	(16.53)	(7.80)	(9.36)	(19.92)
Firm-level controls				
ROA	0.002***	0.000	0.001***	0.002***
	(11.60)	(0.41)	(7.54)	(12.69)
Sales Growth	0.002	0.001*	0.001**	0.004**
	(1.48)	(1.75)	(2.11)	(2.44)
log(*Sales*)	0.012	−0.003	0.004	0.013
	(1.50)	(−1.05)	(1.13)	(1.45)
Leverage	0.000	0.000	0.000	0.000
	(−0.09)	(0.47)	(−0.60)	(0.04)
Country-level controls				
Financial Development	0.001**	−0.0003*	0.001*	0.001**
	(2.08)	(−1.78)	(1.94)	(2.01)
Financial Openness	−0.038***	0.003*	−0.003	−0.038***
	(−7.71)	(1.86)	(−1.32)	(−6.47)
GDP Growth	−0.002***	0.000	0.000	−0.002***
	(−5.21)	(1.24)	(−1.40)	(−4.41)
Institutional quality				
Voice and Accountability	0.000	−0.001	0.002*	0.001
	(0.18)	(−1.02)	(1.82)	(0.27)
Political Stability	0.000	−0.001*	−0.002***	−0.002*
	(0.39)	(−1.73)	(−3.87)	(−1.66)
Government Effectiveness	0.014***	−0.001	0.002*	0.014***
	(5.83)	(−1.13)	(1.83)	(5.24)
Regulatory Quality	−0.019***	0.000	−0.001	−0.020***
	(−8.34)	(−0.49)	(−0.74)	(−7.71)
Rule of Law	−0.003	0.003***	−0.001	−0.001
	(−1.15)	(2.72)	(−0.51)	(−0.17)
Control of Corruption	0.000	0.000	0.000	0.000
	(0.15)	(−0.65)	(0.07)	(0.21)
Country FE	Yes	Yes	Yes	Yes
Industry FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Observations	343,547	343,547	343,547	343,547
R-squared	0.031	0.005	0.01	0.035

Note: Standard errors are clustered at the firm level, and *t*-statistics are reported in brackets. ∗, ∗∗, and ∗∗∗ indicate statistical significance at the 10%, 5%, and 1% level, respectively. Abbreviations: FE, fixed effects; GDP, gross domestic product; NGSC, number of global supply chain relationships; ROA, return on assets.

insignificant difference tests. Therefore, ex ante, there seem to be no significant observable differences in external financing across treated and control firms.

We estimate the following difference-in-differences specification for each of the three financial markets separately, as well as for the aggregation of CBF deals across the three markets:

$$CBF_{it} = \beta_0(Treated_i \times Post_{it}) + \beta_1 Treated_i + \beta_2 Post_{it} + \gamma X_{it} + \delta Z_{kt} + \phi_k + \chi_j + \psi_t + \epsilon_{it}, \quad (2)$$

TABLE 3 Difference-in-differences analysis: Balance tests.

	Control		Treated				
Variables	**Mean**	**Std. Dev.**	**Mean**	**Std. Dev.**	***Diff.***	***t*-stat**	***p*-value**
$\log(Sales_{t-3}+1)$	21.40	3.02	21.42	3.04	−0.009	−0.60	0.551
$\log(Sales_{t-2}+1)$	21.49	3.01	21.52	3.03	−0.011	−0.76	0.445
$\log(Sales_{t-1}+1)$	21.56	3.01	21.60	3.02	−0.013	−0.86	0.389
ROA_{t-3}	0.038	0.124	0.036	0.146	0.011	0.74	0.462
ROA_{t-2}	0.035	0.114	0.033	0.143	0.013	0.89	0.371
ROA_{t-1}	0.035	0.112	0.034	0.139	0.011	0.75	0.452
$Sales\ Growth_{t-3}$	0.124	0.257	0.124	0.310	0.000	−0.02	0.983
$Sales\ Growth_{t-2}$	0.093	0.236	0.099	0.290	−0.020	−1.38	0.167
$Sales\ Growth_{t-1}$	0.086	0.226	0.088	0.279	−0.011	−0.76	0.447
$Leverage_{t-3}$	0.231	0.182	0.226	0.219	0.023	1.56	0.119
$Leverage_{t-2}$	0.232	0.187	0.228	0.232	0.018	1.23	0.217
$Leverage_{t-1}$	0.233	0.190	0.228	0.234	0.024	1.65	0.098
$PreEvent\ 1\{CBF\}$	0.064	0.245	0.064	0.245	0.000	0.00	1.000
Firm number	9433		9433				

Abbreviations: CBF, cross-border financing; ROA, return on assets.

where CBF_{it} is a dummy variable indicating whether firm i in industry j and country k is involved in any CBF deals in year t. $Treated_i$ is a dummy variable set to 1 if firm i has at least one GSC relationship in any year of the sample period. $Post_{it}$ is a dummy variable set to 1 for the treated firm, and its matched control firm if any GSC relationship has been established for the treated firm i by year t. β_0 is the DID regression coefficient of the treatment effect of the GSC relationship on CBF. We control for country, industry, and year fixed effects in all regressions. Furthermore, we include a battery of control variables for firm-level characteristics and country-level attributes such as institutional quality measures, represented by the vector X_{it} and Z_{kt}. All control variables are standardized to have a mean of zero and unit variance.

Table 4 shows the DID results from estimating Equation (2). We report global bond issuance, global equity listing, global loan origination, as well as the overall market. $Prob\{CBF\}$ reports the annual probability of CBF deals for each market for the treated and control firms combined. *Percentage Increase* measures the increase in the probability of using each financial market compared to the annual probability, that is, $Treated_i \times Post_{it}$ divided by $Prob\{CBF\}$. The interaction term, $Treated_i \times Post_{it}$, is significantly positive for each of the individual markets, as well as for CBF deals aggregated across the markets. These findings suggest that firms that have formed at least one GSC relationship have more subsequent CBF deals compared to otherwise similar firms.

Our findings are also economically significant as shown by the estimated coefficients on the interaction term, $Treated_i \times Post_{it}$.[6] For example, when a firm switches from having no GSC relationships to having at least one, there is an increase in the probability of using CBF of 1.7 percentage points. This is a 59% increase relative to the annual probability of the firm obtaining CBF (2.9 percentage points). We further observe that this increase in the likelihood of obtaining CBF is largest in the bond market (1.2 percentage points, a 50% increase), followed by the syndicated loans market (0.4 percentage points, an 87% increase) and the cross-listing market (0.1 percentage points, a 157% increase).

The DID findings are robust along several dimensions. First, since the controls in the DID regressions include industry, country, and time fixed effects, the estimated increase in the probability of using CBF following GSC formation is not driven by any unobservable time-invariant characteristics of industry or the country the treated firm is in or any global trends in GSC and CBF activity. Second, our results are controlled for firm-level characteristics that include firm-level financial factors, as well as variables that measure economic growth (measured as GDP growth), financial development (measured as the ratio of the sum of domestic credit outstanding and equity market capitalization to the GDP), and financial openness (measured by the capital account openness index in Fernández et al., 2016), and institutional quality indices from the World Bank World Governance Indicators (including voice and accountability, political stability, government effectiveness, regulatory quality, the rule of law, and control of corruption).

3.2.2 | Parallel trend tests

The parallel trend assumption is critical for DID analysis. One prediction of the parallel trend assumption is that the difference in obtaining CBF between the treatment and the control groups is constant over time prior to the establishment

TABLE 4 Difference-in-differences analysis: estimation results.

Markets	Global bond issuance (1)	Global equity listing (2)	Global loan origination (3)	Cross-border financing (4)
Treated × Post	0.0121***	0.0011***	0.0040***	0.0171***
	(5.97)	(3.10)	(4.19)	(7.60)
Treated	−0.0011	−0.0002	−0.0006	−0.0020
	(−0.70)	(−0.91)	(−0.87)	(−1.19)
Post	0.0012	−0.0005**	−0.0002	0.0005
	(0.80)	(−2.14)	(−0.23)	(0.27)
Firm-level controls				
ROA	0.0032***	0.0000	0.0015***	0.0046***
	(5.98)	(−0.07)	(4.45)	(6.96)
Sales Growth	0.0101**	−0.0005	0.0000	0.0085*
	(2.48)	(−0.66)	(0.01)	(1.92)
log(Sales)	−0.0287	−0.0015	0.0007	−0.0325
	(−1.04)	(−0.30)	(0.06)	(−1.10)
Leverage	0.0011**	0.0000	0.0005**	0.0015***
	(2.40)	(−0.47)	(2.24)	(3.03)
Country-level controls				
Financial Development	0.0016	−0.0004	0.0008	0.0018
	(0.73)	(−1.56)	(0.62)	(0.69)
Financial Openness	−0.0926***	0.0017	0.0047	−0.0847***
	(−7.47)	(0.49)	(0.77)	(−6.00)
GDP Growth	−0.0045***	0.0001	−0.0009**	−0.0051***
	(−4.58)	(0.26)	(−2.25)	(−4.67)
Institutional quality				
Voice and Accountability	0.0278***	−0.0020	0.0163***	0.0404***
	(4.30)	(−1.36)	(3.75)	(5.18)
Political Stability	0.0036	−0.0012	−0.0010	0.0027
	(0.99)	(−1.24)	(−0.52)	(0.63)
Government Effectiveness	0.0570***	0.0039***	0.0092***	0.0689***
	(8.26)	(3.03)	(3.37)	(9.31)
Regulatory Quality	−0.0420***	−0.0011	−0.0020	−0.0444***
	(−6.60)	(−0.81)	(−0.69)	(−6.32)
Rule of Law	−0.0212***	0.0007	−0.0085**	−0.0274***
	(−2.69)	(0.31)	(−2.01)	(−3.02)
Control of Corruption	0.0004	−0.00192*	−0.0067**	−0.0072
	(0.08)	(−1.67)	(−2.43)	(−1.21)
Country FE	Yes	Yes	Yes	Yes
Industry FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Observations	92,046	92,046	92,046	92,046
R-squared	0.038	0.008	0.010	0.039
Prob{CBF}	0.0240	0.0007	0.0046	0.0288
Percentage Increase	50.42%	157.14%	86.96%	59.38%

Note: Standard errors are clustered at the firm level, and t-statistics are reported in brackets. $*$, $**$, and $***$ indicate statistical significance at the 10%, 5%, and 1% level, respectively. Abbreviations: CBF, cross-border financing; FE, fixed effects; GDP, gross domestic product; ROA, return on assets.

of the GSC relationship. To ensure that the DID estimate is not driven by ex ante trends in the differences in CBF activities prior to GSC formation, we test this prediction of the parallel trends assumption by estimating the following specification:

$$
\begin{aligned}
CBF_{it} = {} & \beta_0(Treated_i \times Post_{it}) + \beta_1(Treated_i \times F2.Post_{it}) \\
& + \beta_2 Treated_i + \beta_3 Post_{it} + \beta_4 F2.Post_{it} \\
& + \gamma X_{it} + \delta Z_{kt} + \phi_k + \chi_j + \psi_t + \epsilon_{it},
\end{aligned}
$$

where $F2.Post_{it}$ is a dummy variable set to 1 for the treatment firm and its matched control firm if any GSC relationship has been established for the treated firm by year $t + 2$. While we have matched the treated and the control firms on a dummy variable indicating whether the firm has been involved in any CBF deal in the 3-year preperiod, this matching method is specifically designed to *not* require the treated and the control firms to be identical in CBF involvement in each of the 3 years. For example, it is possible in our matched sample that the treated firm was involved in a CBF deal 1 year prior to GSC formation, whereas the control firm was involved in a CBF deal 3 years prior to the treated firm's GSC formation event. In other words, we allow these firms to have different trends in CBF in the preperiod. If they do, we would estimate a significant coefficient on $F2.Post_{it}$. Table 5 reports the results that show that the coefficient on $Treated_i \times F2.Post_{it}$ is statistically insignificant. This finding demonstrates that CBF does not increase prior to the formation of the GSC relationship, suggesting that the parallel trends assumption is valid for our results of interest. We also use $F1.Post_{it}$, $F3.Post_{it}$, etc., for alternative specifications and confirm that the parallel trends assumption holds.

We also plot the differences between the treatment and the control groups with confidence intervals for the annual window over the 9-year period starting 3 years before the GSC formation year (year 0) and ending 5 years after. As Figure 2 shows, we cannot reject that the trend of the treated group is the same as that of the control group before the establishment of GSC relationships, meaning that the two groups have similar trends in the probability of CBF for all financial markets. Figure 2 also shows the dynamics of CBF once the GSC relationship is established. The GSC versus non-GSC difference in using CBF is statistically significant for a sustained period of at least 2–5 years across markets following the GSC formation year, suggesting a lasting association between GSC formation and firms' usage of CBF.

To sum up, the parallel trend analysis confirms that our baseline results that firms use more CBF once a GSC relationship is established cannot be purely explained by differential pretrends for GSC versus non-GSC firms. We also observe that the treated firms experience sustained as opposed to temporary increases in the incidences of CBF once the GSC relationships are established.

4 | ROBUSTNESS TESTS

In this section, we conduct several tests to examine the robustness of our main findings. Specifically, we examine whether our findings are robust to alternative measures of CBF as well as to alternative model specifications. We also use exogenous trade agreements as an IV to enhance the causal interpretation of our main findings.

4.1 | Alternative model specification

Since our dependent variable, CBF, is a binary dummy variable, a natural alternative to the linear regression approach we use for our main analysis would be a probit/logit specification. However, the "incidental parameter problem" discussed in, for example, Ai and Norton (2003) suggest that our interaction terms, that is, $Treated_i \times Post_{it}$ as well as the interaction of $Treated_i \times Post_{it}$ with other variables, can be unstable when estimated using nonlinear models such as probit/logit specifications. Previous studies (Barrot, 2016; Duchin & Sosyura, 2014) utilize a linear model to address the concerns raised in Ai and Norton (2003). Therefore, we do not use such nonlinear models as our main model specification. Instead, we run the probit/logit specification here as a robustness check to bolster confidence in our main findings. We report the results of the robustness check in Appendix Table A2 in the Supporting Information, which shows that the main results still remain using the probit/logit approach, that is, the coefficients of GSC in both the logit and probit regressions are significantly positive.

4.2 | Alternative measure of CBF

In our main analysis, our dependent variable CBF is a dummy variable indicating whether firm i in industry j and country k is involved in any CBF deals in year t, which captures an important part of the variation in firms changing usage of CBF. As a robustness test, we redefine CBF as the *number* of deals in each separate market as well as the aggregate number of deals across the three markets. The results of this estimation, reported in panel A of Table A3 in the Supporting Information, are consistent with our earlier findings and suggest that increases in the number of GSC relationships are significantly associated with increases in the number of CBF deals.

As a further test, we also redefine CBF as the *volume* of deals, that is, the exact amount of the CBF deals.[7] The results in panel B of Table A3 in the Supporting Information suggest that an increase in the number of GSC relationships is positively associated with increases in the volume of CBF deals.

TABLE 5 Difference-in-differences analysis: parallel trend tests.

Markets	Global bond issuance (1)	Global equity listing (2)	Global loan origination (3)	Cross-border financing (4)
Treated × Post	0.0116***	0.0012***	0.0037***	0.0165***
	(5.22)	(3.20)	(3.50)	(6.73)
Treated × F2.Post	0.0015	−0.0004	0.0011	0.0019
	(0.63)	(−0.60)	(0.91)	(0.72)
Treated	−0.0021	0.0000	−0.0013	−0.0033
	(−1.06)	(0.01)	(−1.35)	(−1.46)
Post	0.0005	−0.0005	−0.0004	−0.0004
	(0.32)	(−1.62)	(−0.47)	(−0.22)
F2.Post	0.0027	−0.0003	0.0008	0.0032
	(1.47)	(−0.74)	(0.89)	(1.63)
Firm-level controls				
ROA	0.0033***	0.0000	0.0015***	0.0046***
	(6.01)	(−0.08)	(4.46)	(6.98)
Sales Growth	0.0101**	−0.0005	0.0000	0.0084*
	(2.47)	(−0.66)	(0.01)	(1.91)
log(*Sales*)	−0.0289	−0.0015	0.0006	−0.0327
	(−1.04)	(−0.30)	(0.06)	(−1.10)
Leverage	0.0011**	0.0000	0.0005**	0.0015***
	(2.39)	(−0.46)	(2.24)	(3.03)
Country-level controls				
Financial Development	0.0016	−0.0004	0.0008	0.0017
	(0.71)	(−1.54)	(0.60)	(0.67)
Financial Openness	−0.0923***	0.0017	0.0048	−0.0844***
	(−7.44)	(0.48)	(0.79)	(−5.97)
GDP Growth	−0.0045***	0.0001	−0.0009**	−0.0051***
	(−4.57)	(0.26)	(−2.25)	(−4.66)
Institutional quality				
Voice and Accountability	0.0279***	−0.0020	0.0163***	0.0404***
	(4.31)	(−1.36)	(3.76)	(5.19)
Political Stability	0.0035	−0.0012	−0.0011	0.0025
	(0.95)	(−1.22)	(−0.54)	(0.59)
Government Effectiveness	0.0568***	0.0040***	0.0092***	0.0686***
	(8.23)	(3.05)	(3.34)	(9.27)
Regulatory Quality	−0.0426***	−0.0010	−0.0023	−0.0452***
	(−6.68)	(−0.73)	(−0.78)	(−6.42)
Rule of Law	−0.0207***	0.0006	−0.0083**	−0.0268***
	(−2.63)	(0.27)	(−1.97)	(−2.95)
Control of Corruption	0.0003	−0.0019*	−0.0068**	−0.0074
	(0.05)	(−1.65)	(−2.45)	(−1.24)
Country FE	Yes	Yes	Yes	Yes
Industry FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Observations	92,046	92,046	92,046	92,046
R-squared	0.038	0.008	0.010	0.039

Note: Standard errors are clustered at the firm level, and *t*-statistics are reported in brackets. ∗, ∗∗, and ∗∗∗ indicate statistical significance at the 10%, 5%, and 1% level, respectively. Abbreviations: FE, fixed effects; GDP, gross domestic product; ROA, return on assets.

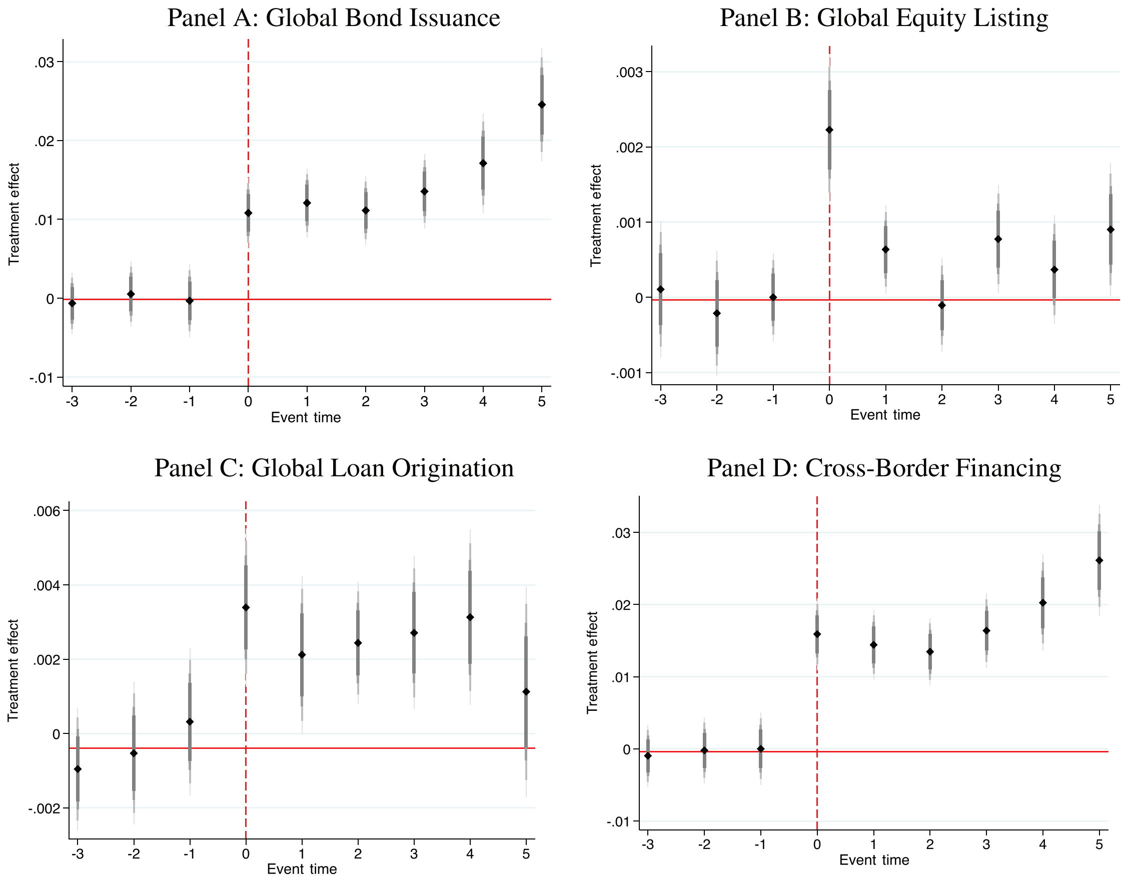

FIGURE 2 Difference-in-differences analysis: parallel trend plots. [Color figure can be viewed at wileyonlinelibrary.com]

4.3 | Instrumental variable

While we employ a battery of control variables and match firm-level characteristics that may affect CBF in the DID analysis, there may be other unobservable factors or unknown missing factors that could affect our analysis. To mitigate the potential endogeneity concern caused by unobservable or missing factors that may obscure our main results, we use an IV approach to conduct a two-step least squares estimation.

To construct our IV, we collected data on the introduction of new trade agreements at the country and year level using the Trade Analysis Information System (TRAINS) data from the World Bank. We compute the logarithm of the number of trade agreements reported in TRAINS, $\log(1 + NTradeAgreement_{k,t})$, for each country in each year. Over time, variation in this variable reflects new trade agreements that a country enters into. For firms in the country, we use this variable as an instrument for their respective log number of GSC relationships.

The instrument we use satisfies the relevance condition since international trade agreements lower the costs of doing business globally, including the establishment of new GSC relationships. Empirically, column 1 of Appendix Table A4 in the Supporting Information shows that the instrument positively predicts $\log(1 + NGSC_{it})$ with a t-stat of 9.68, suggesting that the relevance condition is satisfied. To ascertain whether trade agreements are related to other country-level factors relevant to CBF, that is, the exclusion restriction, we perform a test by regressing other relevant country-level factors on the trade agreement instrument. The results reported in Table A5 in the Supporting Information suggest that other factors potentially associated with CBF (financial development and openness, GDP growth, and the various institutional variables) and the instrument are uncorrelated.

Our IV analysis is then based on estimating the following Two-Stage least squares (2SLS) specification:

$$\log(1 + NGSC_{it}) = \alpha \log(1 + NTradeAgreement_{k,t-1}) + \gamma X_{it} + \delta Z_{kt} + \phi_k + \chi_j + \psi_t + \epsilon_{it}$$

$$CBF_{it} = \beta \log(1 \widehat{+ NGSC_{it}}) + \gamma X_{it} + \delta Z_{kt} + \phi_k + \chi_j + \psi_t + \epsilon_{it}$$

where $NGSC_{it}$ is the number of GSC relationships firm i has in year t, $NTradeAgreement_{k,t-1}$ is the number of new trade agreements for each country k in year $t-1$, CBF_{it} is a dummy variable indicating whether firm i in country k is involved in any CBF deals in year t. We also control

TABLE 6 Cross-sectional analyses.

Dependent variable: Cross-border financing	Small versus large firm (1)	High versus low institutional ownership (2)	More versus less analysts following (3)
Treated × Post	0.0186***	0.0194***	0.0193***
	(7.31)	(7.37)	(7.60)
Treated × Post × Large/High/More	−0.0098*	−0.0152***	−0.0134**
	(−1.83)	(−2.81)	(−2.30)
Treated	−0.0008	−0.0036*	−0.0022
	(−0.43)	(−1.94)	(−1.20)
Post	0.0001	0.0030	0.0028
	(0.04)	(1.50)	(1.45)
Treated × Large/High/More	−0.0048	0.0077	0.0023
	(−1.15)	(1.54)	(0.43)
Post × Large/High/More	−0.0009	−0.0083**	−0.0114***
	(−0.23)	(−2.38)	(−2.84)
Large/High/More	−0.0083**	0.0031	
	(8.83)	(−2.52)	(0.82)
Firm-level controls			
ROA	0.0043***	0.0051***	0.0050***
	(6.53)	(7.56)	(7.37)
Sales Growth	0.0086*	0.0086*	0.0085*
	(1.94)	(1.93)	(1.92)
log(Sales)	−0.0397	−0.0350	−0.0354
	(−1.34)	(−1.18)	(−1.19)
Leverage	0.0015***	0.0015***	0.0015***
	(3.01)	(3.03)	(2.99)
Country-level controls			
Financial Development	0.0023	0.0021	0.0022
	(0.92)	(0.83)	(0.84)
Financial Openness	−0.0868***	−0.0859***	−0.0861***
	(−6.11)	(−6.08)	(−6.10)
GDP Growth	−0.0054***	−0.0050***	−0.0050***
	(−4.91)	(−4.54)	(−4.47)
Institutional quality			
Voice and Accountability	0.0405***	0.0408***	0.0409***
	(5.18)	(5.23)	(5.24)
Political Stability	0.0027	0.0037	0.0043
	(0.63)	(0.88)	(1.01)
Government Effectiveness	0.0703***	0.0662***	0.0664***
	(9.47)	(8.94)	(8.96)
Regulatory Quality	−0.0435***	−0.0474***	−0.0477***
	(−6.19)	(−6.71)	(−6.73)
Rule of Law	−0.0269***	−0.0270***	−0.0275***
	(−2.96)	(−2.98)	(−3.04)
Control of Corruption	−0.0048	−0.0068	−0.0070
	(−0.80)	(−1.13)	(−1.17)

(Continues)

TABLE 6 (Continued)

Dependent variable: Cross-border financing	Small versus large firm (1)	High versus low institutional ownership (2)	More versus less analysts following (3)
Country FE	Yes	Yes	Yes
Industry FE	Yes	Yes	Yes
Year FE	Yes	Yes	Yes
Observations	92,046	92,046	92,046
R-squared	0.043	0.040	0.039

Note: Standard errors are clustered at the firm level, and *t*-statistics are reported in brackets. *, **, and *** indicate statistical significance at the 10%, 5%, and 1% level, respectively.
Abbreviations: FE, fixed effects; GDP, gross domestic product; ROA, return on assets.

for firm-level characteristics X_{it}, country-level attributes and institutional quality measures Z_{kt}, and country, industry, and time fixed effects ϕ_k, χ_j, and ψ_t.

The results of this analysis, reported in Table A4, are consistent with the causal interpretation that the formation of GSC relationships leads to CBF. For example, for the CBF covering the full set of financing markets (reported in column (5)), the IV estimate is positive and significant at the 1% level. It implies that going from 0 to 1 GSC relationship is associated with an increase of 5.5% in the probability of CBF. The IV results are also statistically significant in two of the three specific financing markets we study.

5 | CROSS-SECTIONAL ANALYSES

In this section, we investigate the cross-sectional nature of our sample to further characterize the effect of establishing new GSC relationships on subsequent CBF to bolster support for the validity of our conclusions. That is, the cross-sectional tests in this section are aimed, in part, at narrowing the range of potential alternative explanations for our findings. We consider cross-sectional characteristics that are expected to generate differing effects of GSC formation on subsequent CBF.

We begin with the observation that, theoretically, the effect of establishing new GSC relationships on reducing information obstacles to obtaining CBF is not expected to be uniform across firms. More specifically, for firms that are already highly visible, the marginal benefit in obtaining CBF associated with the increased visibility due to establishing a new GSC relationship is limited. Therefore, we conjecture that the observed increase in the usage of CBF once a GSC relationship is established should be higher for firms with lower visibility, while the increase in CBF following GSC integration should be lower for more visible firms.

To capture the heterogeneity in firm visibility, we rely on three common proxies used in the extant literature. The first proxy we use is firm size (Bushee & Miller, 2012). Larger firms are more visible in the economy and generate more public information in the market (Baloria & Heese, 2018).

Second, we use institutional ownership to proxy for the visibility of a firm (Han et al., 2017; Hou & Moskowitz, 2005). Extant evidence and simple intuition suggest that firms with larger institutional holdings are typically more visible. For example, previous research has found that information generated about a firm increases with institutional ownership and that institutional owners tend to demand greater information transparency (Falkenstein, 1996; O'Brien & Bhushan, 1990). Institution ownership also correlates with firm size. Kaniel et al. (2012) show that larger firms also attract more institutional investor awareness.

Third, we use the number of analysts following the firm. Prior research has shown that greater analyst following leads to more information being gathered and disseminated about the firm (Chakravarty et al., 2022; O'Brien & Bhushan, 1990; O'Brien & Tan, 2015). Thus, in acting as information intermediaries, analysts directly affect firm visibility (Brennan & Subrahmanyam, 1995; Han et al., 2017). Previous research shows that analyst coverage correlates with firm size and institutional ownership. Bushee and Miller (2012) suggest that large firms attract high analyst coverage due to the economics of analyst research. Falkenstein (1996) illustrates that institutional investors such as mutual funds, pensions, and money managers prefer to invest in firms that have a sizable analyst following due to the transparent information environment.

For our cross-sectional tests, we create an indicator variable for each of our proxy measures, which takes the value of 1 (indicating more visible) if a firm is ranked among the top half in terms of firm size ($Large_i$), the percentage of institutional ownership ($High_i$), or the number of analysts following ($More_i$). Then, we use the indicator to further interact with the DID term $Treated_i \times Post_{it}$ to identify the incremental effect of firm visibility on the likelihood of subsequent CBF. Note that our DID specification is a linear probability model.[8] Therefore, the coefficient on the $Indicator_i \times Treated_i \times Post_{it}$ triple interaction term measures the percentage point difference in the probability of using CBF between the high visibility firms and the rest of the sample firms.

The results, reported in Table 6, show that there is a significantly negative incremental effect of GSC formation on CBF for firms that are large in size, held more by institutional investors, and followed by more analysts. Following GSC integration, large firms experience an approximately 53% decrease (0.0098 divided by 0.0186) in the probability of using CBF than small firms. Firms with greater institutional ownership and firms with more analyst coverage suffer a 78% decrease and a 69% decrease, respectively. These findings indicate that the observed increase in CBF following the establishment of a GSC relationship is concentrated in those firms with lower visibility. This robustly observed pattern is consistent with an information channel, whereby establishing new GSC relationships lowers information obstacles to obtaining CBF.

Finally, we recognize that while improved informational visibility is helpful for firms overcoming obstacles to obtaining CBF, firms may also choose CBF due to operational needs that arise from establishing new GSC relationships.[9] Therefore, following the establishment of a new GSC relationship, the increased usage of CBF, which reflects more observed accessibility, is a joint outcome of accessibility boosted by information visibility and operational needs.

6 | CONCLUSION

In this paper, we provide evidence that the formation of GSC partnerships increases a firm's subsequent usage of CBF. Our findings are evident in all three major financing markets—equities, syndicated loans, and public debt. We show that the relationship between GSC formation and CBF is robust to controls for standard firm-level determinants as well as for industry- and country-specific shocks. In addition, a difference-in-differences model with attribute matching supports a causal interpretation of this empirical relationship, that is, integration into the GSC network increases capital raising in global financial markets by supply chain partners. Furthermore, our heterogeneity analysis indicates that less visible firms, in terms of smaller size, less institution ownership, as well as less number of analysts following, can benefit more through integrating into the GSC network.

Our paper represents a first step in considering the impact of global supply chain integration on the usage of global financial markets. While supply chain finance has been widely studied in the Operation Management (OM) literature, few papers study how supply chain formations affect traditional capital raising in formal financial markets (external to the supply chain relationship). Our focus on the causal relationship between GSC and CBF bridges this gap. One question that remains is whether CBF will have economic benefits for firms with GSCs. When a firm's financial market improves, the firm's business may grow together with an improvement in productivity (Bai et al., 2018). Other potential benefits include diversification of capital sources, decreasing exchange rate risk exposure, certification benefits, as well as other resources obtained via interactions with global financing institutions. There may also be an amplification effect, where greater usage of CBF leads domestic financiers to provide better terms due to increased competition. However, other research argues that domestic opportunities significantly reduce the net benefits of foreign financing for some firms (Leuz & Oberholzer-Gee, 2006). It is possible that global financing imposes additional costs on companies, as companies cross-listing on overseas exchanges need to adapt to the country's regulations (Reese & Weisbach, 2002). More financing opportunities also bring greater challenges to firms. In sum, quantifying these subsequent benefits and challenges of CBF for firms with GSC would be a meaningful future exercise. We leave these further investigations to future research.

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ENDNOTES

[1] Specifically, we use a Python package (*cleanco*) to remove suffixes and special symbols in company names. Then we use the fuzzy character string matching algorithm to relate company names from the three CBF datasets with Worldscope company names respectively. We pay particular attention to the names of parent firms and their multiple subsidiary firms. For companies that do not result in an exact match, we perform a manual match with groups of two research assistants who independently verify each inexact match using information from Bloomberg queries and web searches to make judgments. We review them again using a third research assistant if different judgments occur.

[2] Some firms conducted more than one CBF deals in the year; the median number of CBF deals in these firm-year observations is 1, whereas the mean number of CBF deals is 1.82.

[3] The logarithm transformation also facilitates the economic interpretation as it gives the percentage increase (or decrease) in the response for every one-unit increase in the independent variable. We also take the logarithm of *Sales* in the control variable to reduce its range and the estimation impact of observations with outlying sales.

[4] For details, refer to https://mba.tuck.dartmouth.edu/pages/faculty/ken.french/Data_Library/det_12_ind_port.html.

[5] Same as before, some firms conducted more than one CBF deals in the year. The median number of CBF deals in these firm-year observations is 1, and the mean number of CBF deals is 1.70.

[6] In calculating economic significance, we note that our specification is a linear probability model, such that the coefficient on the $Treated_i \times Post_{it}$ term is the marginal effect of CBF increase after establishing GSCs.

[7] In measuring volume, it is difficult to determine the exact amount of funding from global equity listings, and there is also missing information on the exact amount of CBF deals elsewhere in our dataset. Nevertheless, we conduct a robustness test to our best effort by collecting as much information as possible on the volume of CBF deals. The volume of bonds and loans comes from Thomson Reuters Datastream and Thomson Reuters DealScan, respectively. The amount of global equity listing comes from the increased amount of common equity that year.

[8]The linear probability model is more suited for estimating specifications with higher order interaction terms compared to the logit and the probit models (Ai & Norton, 2003).

[9]One such operational need is to hedge exchange rate risk (Kazaz et al., 2005). In Appendix Table A6 in the Supporting Information, we examine the exchange rate volatility and find significant (at 10% level) and positive results for the overall CBF market.

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SUPPORTING INFORMATION

Additional supporting information can be found online in the Supporting Information section at the end of this article.

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